SmartDesk AI
Automated Ticket Classification & Routing

Presented by
R Manju Bhargav
S Saniya

INTRODUCTION

What is Customer Support Ticket Classification?

Customer support ticket classification is the process of automatically sorting and labeling incoming customer or user issues into predefined categories based on the content of their request.

Why It Is Needed:

Reduces manual work for support teams

Ensures tickets reach the correct team faster

Helps prioritize urgent issues

Ensures consistent work without human error

What the Base Research Did

Base Paper Summary:

- Used Google Vertex AutoML
- Small Jira dataset(1000 rows)
- AutoML did the feature engineering and training
- No deployment
- No automation or routing
- Only focused on the classification model results

What is AutoML?

AutoML (Automated Machine Learning) is a system that automatically builds machine learning models without requiring the user to manually perform tasks like data preprocessing, feature engineering, model selection, or hyperparameter tuning.

You give the data

AutoML tries multiple algorithms

Picks the best model for you.

What My Project Does

My Enhancements:

- Used **BERT** transformer for deeper text understanding and improved accuracy
- Large,diverse dataset (11k rows)
- Added **n8n** workflow for automated routing
- Send categorized outputs to Telegram support teams

Comparison Table

BASE RESEARCH		MY PROJECT
AutoML (Vertex AI)	MODEL	BERT
AutoML handled	PREPROCESSING	Custom cleaning + lemmatization
None	DEPLOYMENT	Flask API

BASE RESEARCH		MY PROJECT
Limited	PRACTICAL USE	Fully Deployable
No	AUTOMATION	n8n Workflow

System Workflow Comparison

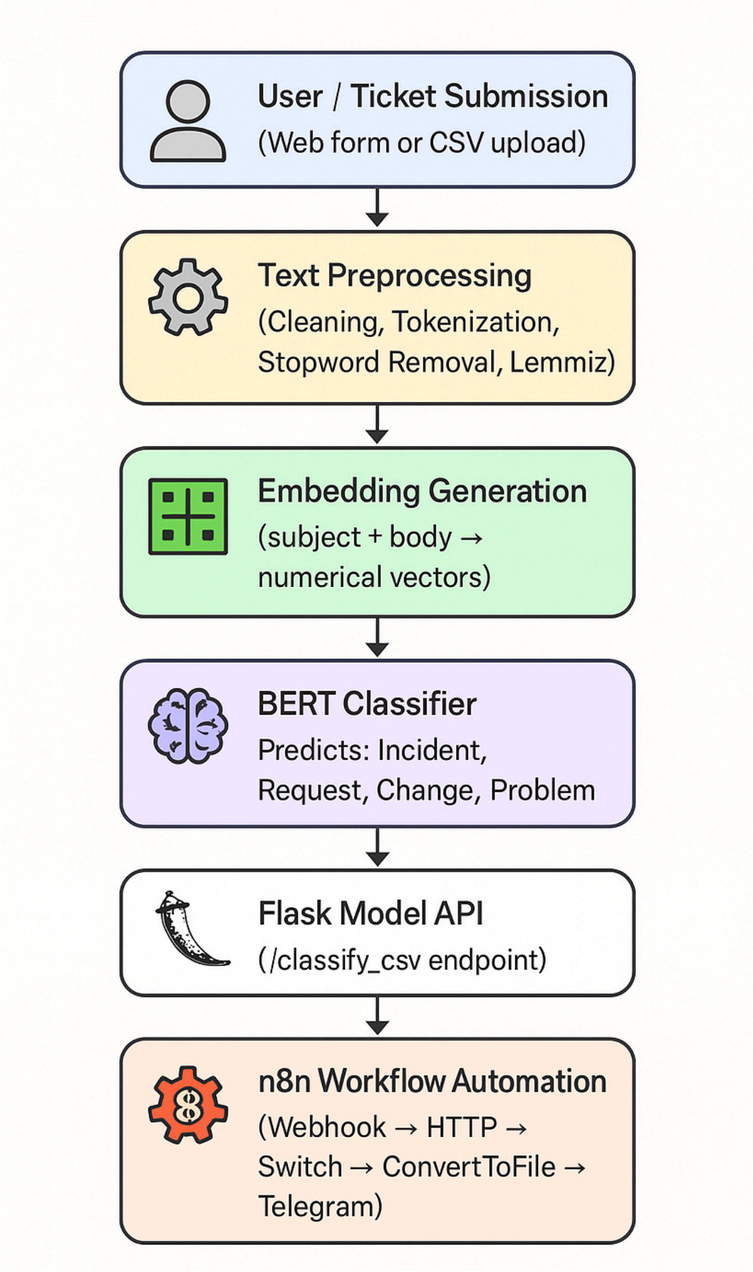
Base Paper Workflow:

Dataset

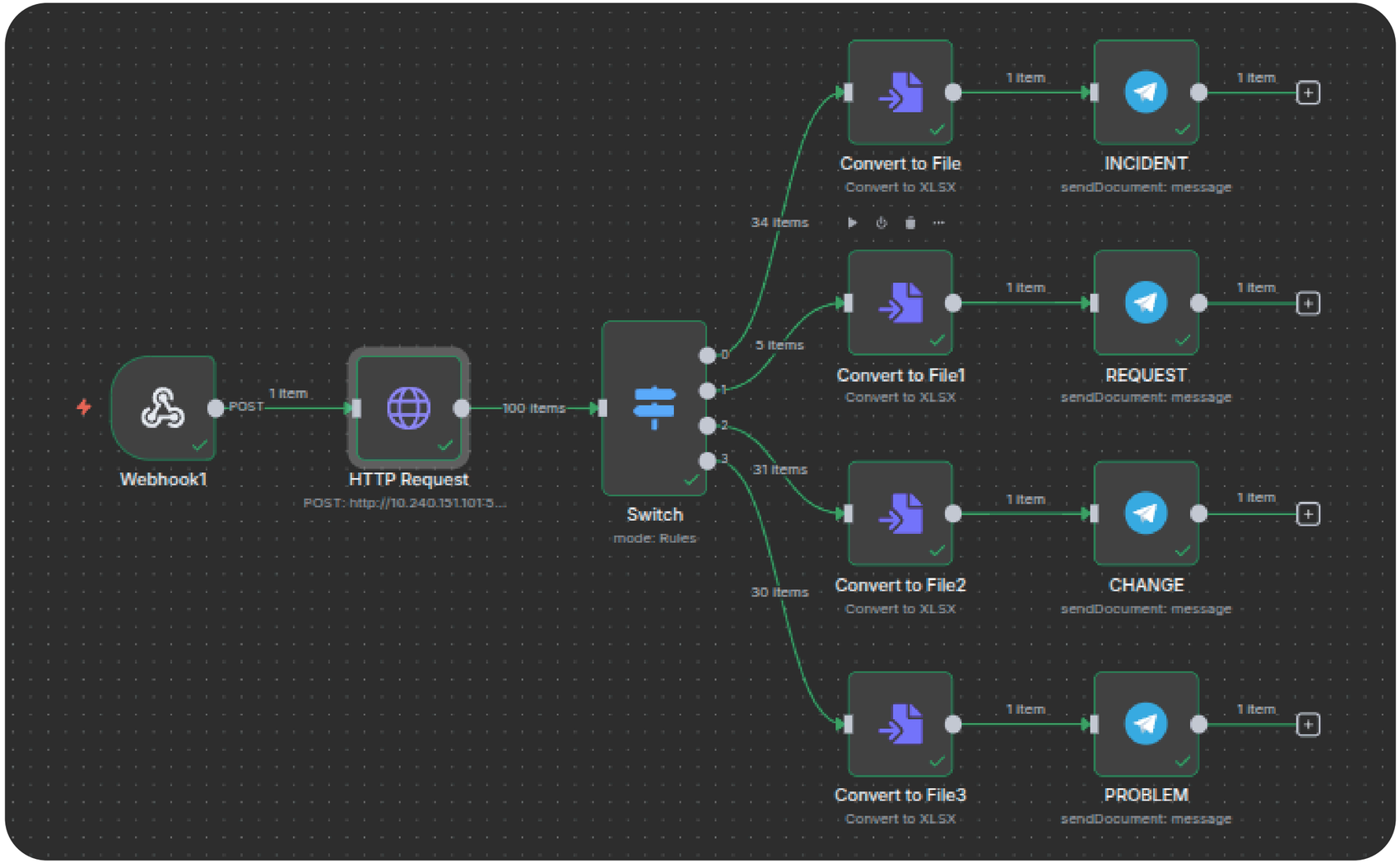
Google Vertex AI (AutoML)

Metrics

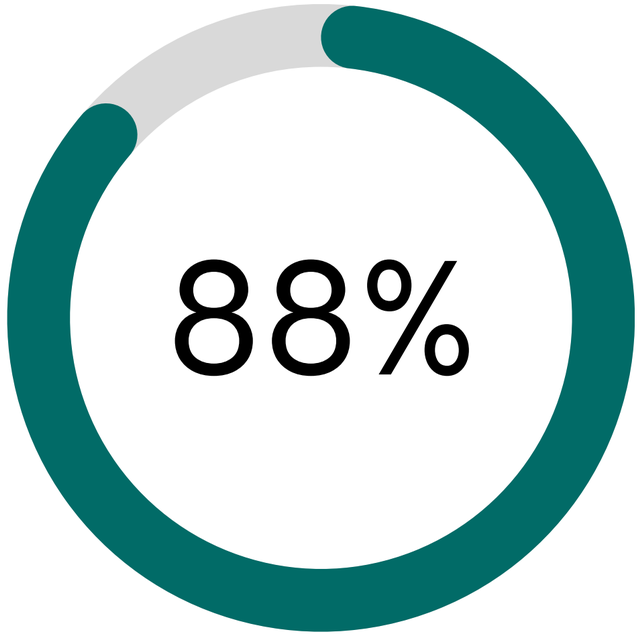
My Workflow:


User / Ticket Submission
(Web form or CSV upload)
Text Preprocessing
(Cleaning, Tokenization, Stopword Removal, Lemmiz)
Embedding Generation
(subject + body → numerical vectors)
BERT Classifier
Predicts: Incident, Request, Change, Problem
Flask Model API
(/classify_csv endpoint)
n8n Workflow Automation
(Webhook → HTTP → Switch → ConvertToFile → Telegram)

My Project n8n workflow

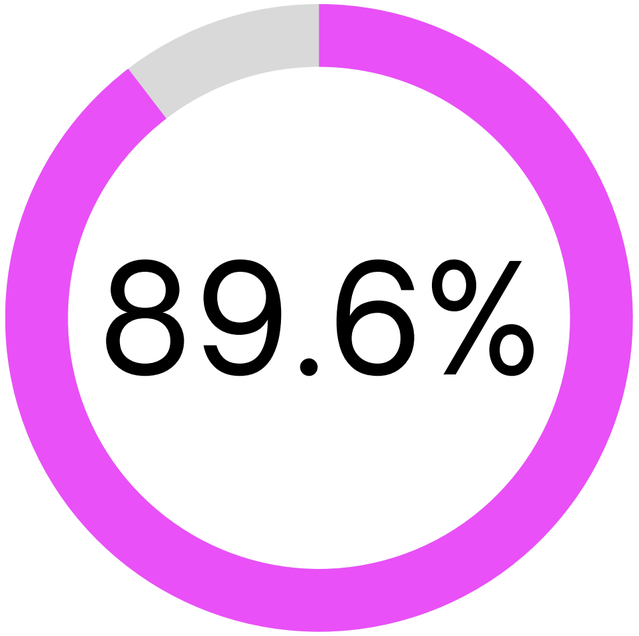


Accuracy Comparison



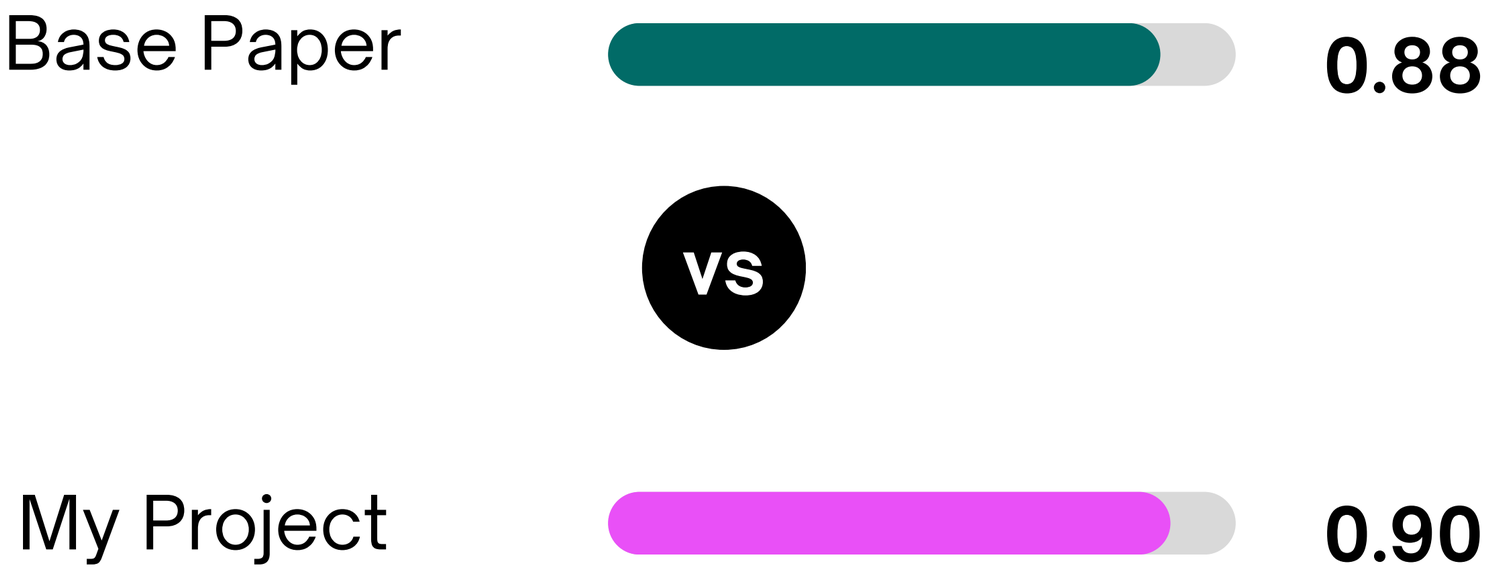
Base Paper

vs



My Project

F1-Score Comparison



Improvements I Made

Moved from AutoML → Contextual BERT model

Introduced custom preprocessing

Added real Flask API deployment

Built full workflow automation

Implemented real-time routing to Telegram

Created a complete end-to-end ITSM solution (not just a model)

Conclusion

- Base research focused on model accuracy only
- My work focuses on real-world usage
- Combines Deep Learning + Automation
- More practical, scalable, deployable
- Direct impact on customer experience and IT team efficiency

References

M. Truss and S. Böhm, “AI-based Classification of Customer Support Tickets: State of the Art and Implementation with AutoML,” IWEMB, 2021.

Base Paper: https://arxiv.org/abs/2406.01789

J. Devlin, M.-W. Chang, K. Lee and K. Toutanova,
“BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” NAACL, 2019.

Thank you!